Career Roadmap & Certification Guide

Trainee System Engineer to International-Level Infrastructure Engineer

Part 1 — Career Roadmap (12-month plan)

Part 2 — Certification Roadmap

Part 1: Career Roadmap

Trainee System Engineer → International-Level Infrastructure Engineer

Your goal should be to grow from a Trainee System Engineer at Premier System Ltd., Islamabad, into an engineer who can confidently design, deploy, troubleshoot, and manage enterprise infrastructure for international employers.

Since your company works mainly with Red Hat OpenShift and VMware, the smartest strategy is to build strong infrastructure fundamentals first, then specialize in virtualization, Linux, cloud-native platforms, and automation.

My recommendation is to follow a 12-month roadmap, with measurable skills and real projects—not just certificates.

Your target: enterprise infrastructure, virtualization, and cloud-native engineering.

1. Your career destination

Stage 1 — Infrastructure Engineer

Months 1–3

Learn servers, storage, networking, Linux, hardware management, and troubleshooting.

Stage 2 — Virtualization & Linux Engineer

Months 4–6

Develop practical skills in VMware vSphere, Red Hat Enterprise Linux, and high availability.

Stage 3 — Automation & OpenShift Engineer

Months 7–9

Build Kubernetes/OpenShift skills, scripting, Ansible automation, and deployment workflows.

Stage 4 — International Job Readiness

Months 10–12

Build a technical portfolio, demonstrate production-style projects, and prepare for international interviews.

These are learning stages, not guaranteed promotion dates. Your progress will depend on hands-on experience, the responsibilities you receive at work, and how consistently you practise.

2. Your 12-month learning plan

Months 1–2: Server and data-center fundamentals

FIRST PRIORITY

Learn:

- CPU, RAM, motherboard, BIOS/UEFI and firmware.
- HPE iLO and Dell iDRAC for remote server management.
- HDD vs SSD vs NVMe; RAID 0, 1, 5, 6, 10.
- SAN vs NAS, Fibre Channel, iSCSI, LUNs and multipathing.
- Server power, rack units, cooling, cabling and redundant PSUs.

Practical output: Draw an enterprise server connectivity diagram showing servers, management ports, switches, storage and power.

Month 3: Enterprise networking

Learn:

- IP addressing, subnetting, DNS, DHCP and routing.
- VLANs, trunk ports, LACP and spanning tree.
- TCP/UDP, firewalls, MTU and troubleshooting.
- Management, storage and production network separation.

Practical output: Create a network diagram with a management network, server network and redundant storage paths.

Months 4–5: Linux administration

Focus on Red Hat Enterprise Linux (RHEL).

- Users, groups, permissions, processes and services.
- SSH, systemd, logs, storage partitions and filesystems.
- LVM, mounts, firewall rules and SELinux basics.
- Package management, networking and system recovery.

Practical output: Install a Linux VM, configure SSH and a web service, add a disk, investigate service failures and document recovery.

Month 6: VMware virtualization

Learn:

- Hypervisors, ESXi hosts and vCenter.
- Virtual machines, virtual switches and port groups.
- Datastores, VMFS and shared storage.
- HA, vMotion, resource allocation and snapshots.
- Backup, recovery and cluster troubleshooting.

Practical output: Build a small virtual lab and document how you would recover a VM after a host failure. Practise migration only where your lab supports it.

Months 7–9: OpenShift and automation

Learn:

- Containers, images, Kubernetes and YAML.
- Pods, deployments, services, routes and persistent volumes.
- OpenShift projects, RBAC, secrets and resource limits.
- Ansible playbooks, inventory, variables and templates.
- Bash scripting, Git and basic CI/CD concepts.

Practical output: Deploy a containerized application, expose it through a route, configure storage, and automate a repeatable Linux configuration task.

Months 10–12: Cloud, security and job readiness

Learn:

- AWS fundamentals: IAM, VPC, EC2, S3 and CloudWatch.
- Infrastructure as Code fundamentals with Terraform.
- Monitoring, patching, backups and incident response.
- GitHub portfolio, technical documentation and interview skills.

Practical output: Publish a complete infrastructure project with an architecture diagram, configuration files, deployment steps, troubleshooting notes and a recovery plan.

3. Which skills should you prioritize?

For your specific job, I would rank them like this:

Priority	Skill	Why it matters
1	Server, storage and networking	Foundation for almost every infrastructure task
2	Linux and troubleshooting	Essential for enterprise systems and OpenShift
3	VMware virtualization	Directly relevant to your workplace
4	OpenShift and Kubernetes	Your key specialization opportunity
5	Ansible and scripting	Helps you automate repetitive work
6	AWS and Terraform	Extends your skills toward cloud and platform engineering

Important: Don't try to master all six at once. Learn the fundamentals, then connect each new concept to something you encounter at work.

4. Certifications: spend strategically

International employers value practical ability, experience and credible certifications together. You do not need to purchase every certification.

1. RHCSA — Red Hat Certified System Administrator

FIRST MAJOR TARGET

This is a strong foundation for your Linux and OpenShift path. Start with the RH124 learning material, then RH134-level administration and hands-on exam preparation. The official exam is EX200.

Official resource: Red Hat training and certifications

2. OpenShift Administrator

Once your Linux foundation is solid, progress toward DO180, then DO280 and the EX280 certification as your practical skills develop. Follow the official current learning path rather than booking an exam too early.

Official resource: OpenShift skills paths

3. VMware certification or training

Choose a current VMware Cloud Foundation/vSphere learning path that matches the systems your employer actually deploys. Product names, course paths and certification requirements change, so verify the current route before paying.

Official resource: Broadcom VMware Education

4. Ansible automation

Learn Ansible basics and YAML first. Later, consider RH294-level training and the relevant current automation certification.

Official resource: Free Ansible learning resources

My recommendation: begin with free official resources and your company's lab environment. Pay for an exam only when you can perform the tasks independently and understand the exam objectives.

5. Your daily routine while working and studying your MS

Since you are working full-time and pursuing your MS, your roadmap must be sustainable. You don't need to study four hours every day.

Weekly learning schedule

When	What
At work	Learn from real tasks: ask questions, study approved diagrams, observe deployments, record troubleshooting steps and review official product documentation.
Monday–Thursday — 45 min/day	Study one concept and practise it in a lab. Prefer hands-on practice over watching long playlists.
Friday — 30 min	Review the week's notes, commands, mistakes and troubleshooting lessons.
Weekend — 2 hours	Complete one practical lab, update your portfolio and plan the next week's learning.
Total	Approximately 5–6 hours per week outside work.

Protect time for your MS thesis and rest. Consistent learning over a year is more valuable than an exhausting schedule that lasts only a few weeks.

6. How to grow faster inside Premier System Ltd.

This is where you can gain an advantage over people who only study courses.

Follow the 5-step workplace growth method

1. **Understand every assigned task.** Before making changes, understand what the component does, why it is required and how it connects to the rest of the infrastructure.
2. **Maintain a private learning log.** Record the problem, symptoms, root cause, solution and lessons learned. Never include customer credentials, confidential configurations or restricted company information.
3. **Request supervised hands-on work.** Progress from observing server installations and network diagrams to assisting with configuration, validation and troubleshooting.
4. **Learn to produce professional deliverables.** Practise connectivity diagrams, BoQs, compliance matrices, implementation plans, test reports and as-built documentation.
5. **Become reliable before trying to look advanced.** Verify specifications, follow change procedures, document risks and escalate when a change could affect production.

For example, your HPE server and connectivity work can help you learn how CPU, memory, RAID controllers, network adapters, management interfaces and storage fabrics fit together. HPE's official iLO documentation is a useful reference for server management and monitoring.

Your goal should be to become the trainee who can explain why a solution works, not just copy a configuration from someone else.

7. Build a portfolio that international employers can evaluate

By the end of 12 months, aim to have these four projects completed and documented.

Project 1: Enterprise infrastructure design

Draw a server rack and connectivity architecture, including redundant switches, management networking, storage connectivity, power and a bill of materials.

Project 2: Linux server deployment

Deploy a Linux server, configure users and SSH, host a service, configure a firewall, monitor logs and write a recovery procedure.

Project 3: OpenShift application platform

Deploy an application, configure a service and route, apply resource limits, configure persistent storage where available and document troubleshooting.

Project 4: Infrastructure automation

Use Ansible to configure multiple Linux VMs consistently. Store your playbooks in Git and include a test showing that repeated runs produce the intended configuration.

Publish sanitized project files and documentation on GitHub. Do not upload Premier System's customer documents, internal diagrams, BoQs or confidential information.

8. Where this roadmap can take you

After building sufficient practical experience, you could target roles such as:

- Linux System Administrator / Linux Engineer
- VMware / Virtualization Engineer
- Infrastructure Engineer
- Red Hat OpenShift Administrator
- Platform Engineer
- Cloud Infrastructure Engineer

Senior engineer and architect roles typically require deeper production experience, system design, incident handling and responsibility for complex deployments. Treat them as longer-term targets, not titles that a 12-month study plan automatically guarantees.

9. Your first 30 days: start here

First-month checklist

Week 1: Server hardware

- ☐ Understand CPU, RAM, DIMMs, PCIe and power supplies
- ☐ Learn HPE iLO and Dell iDRAC fundamentals
- ☐ Draw a basic physical server diagram

Week 2: Storage

- ☐ Understand HDD, SSD, NVMe and RAID
- ☐ Learn SAN, NAS, iSCSI and Fibre Channel
- ☐ Draw server-to-storage connectivity

Week 3: Networking

- ☐ Practise IP addressing and subnetting
- ☐ Understand VLANs, gateways, DNS and DHCP
- ☐ Draw a management and production network

Week 4: Linux and review

- ☐ Install a Linux VM
- ☐ Practise users, SSH, services and logs
- ☐ Complete one troubleshooting report

My final advice

Your fastest sustainable path is:

Infrastructure fundamentals → Linux → VMware → OpenShift → Ansible → Cloud.

Don't chase every trending technology or collect certificates without practical experience. Build one strong foundation, demonstrate it through real tasks and projects, and then move to the next level. Your existing workplace is an opportunity to connect what you learn with enterprise systems.

Part 2: Certification Roadmap

Certification roadmap for your career

For your Trainee System Engineer role at Premier System Ltd., I recommend focusing on Red Hat, OpenShift, VMware, and Ansible first. Add AWS later to broaden your international opportunities.

The certifications I recommend, in order

1. RHCSA — Red Hat Certified System Administrator

START HERE · TARGET: MONTHS 1–4

Your foundation for Linux administration, troubleshooting, storage and services. This is my first choice for you.

Official certification details

2. Red Hat OpenShift administration

TARGET: MONTHS 5–9

Progress toward the current OpenShift administrator credential and its EX280 exam after building Linux and Kubernetes fundamentals. Red Hat updated its certification framework in May 2026, so confirm the current credential title and requirements before registering.

Official Red Hat certification paths

3. VMware vSphere / VCF certification

TARGET: MONTHS 5–10

Choose the current VMware Cloud Foundation or vSphere path that matches your company's deployed version. Prioritize ESXi, vCenter, clusters, storage, networking and HA before the exam.

Official VMware certification information

4. Ansible automation

TARGET: MONTHS 8–11

Learn playbooks, YAML, inventories and automated Linux configuration. Once comfortable, consider the RHCE/EX294 route or the relevant current Ansible credential.

Official Red Hat training

5. AWS Certified Solutions Architect – Associate

TARGET: MONTHS 10–12 OR LATER

Add AWS after developing a strong infrastructure foundation. This certification covers cloud architecture and expands your options beyond on-premises infrastructure.

Official AWS certification page

Which certifications should you pay for first?

Certification	My recommendation
RHCSA	First paid certification
OpenShift administrator	Second priority
VMware VCF/vSphere	Pursue if relevant to your work
Ansible / RHCE	After Linux fundamentals
AWS Solutions Architect Associate	Later, for cloud expansion
CompTIA A+ / Server+	Optional; lower priority for your current enterprise role

Your first 90 days

- **Month 1:** Study RHCSA topics and practise Linux in a virtual machine.
- **Month 2:** Continue RHCSA hands-on practice and learn networking and storage.
- **Month 3:** Test yourself against the official RHCSA objectives and decide whether you're ready for the exam.

You can use Red Hat's official training information and student access options to explore learning resources before paying for an exam.

One important question: Does Premier System Ltd. provide you with access to official Red Hat and VMware training or certification vouchers, or will you pay for certifications yourself? That will help me prioritize the most cost-effective path for you.